‹#›



1) Factorise these quadratic expressions:

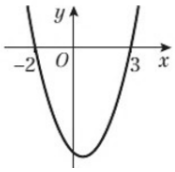
a) $x^2 + 6x + 5$ $(x + 5)(x + 1)$

b) $x^2 - 4x + 3$ $(x - 3)(x - 1)$

2) Sketch the graphs of the following functions:

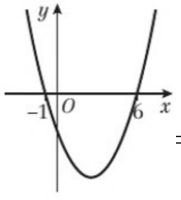
a) $y = (x + 2)(x - 3)$

$(0, -6), (-2, 0), (3, 0)$ Min at $(0.5, -6.25)$

a 

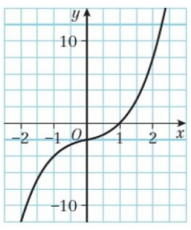

b) $y = x^2 - 6x - 7$

$(0, -7), (-1, 0), (7, 0)$ Min at $(3, -16)$

b 


3) Copy and complete the table of values for t $= x^3 + x - 2$.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | -4 | -2.625 | | | 0 | 2.875 | 8 |



**1a**: Cubic Graphs

**1b**: Quartic Graphs

**1c**: Reciprocal Graphs

**2**: Points of Intersection

**3**: Graph Transformations



In Chapter 2 we briefly saw that a **polynomial** expression is of the form:

$$a + bx + cx^2 + dx^3 + ex^3 + \dots$$

where $a, b, c, d, e, \dots$ are constants (which could be 0).

The **order** of a polynomial is its highest power.

| Order | Name |
|---|---|
| 0 | Constant (e.g. "4") |
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |

These are covered in Chapter 5.

Chapter 2 explored the graphs for these.

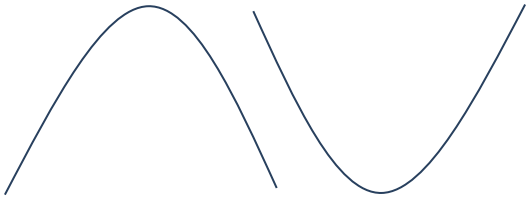
We will cover these now.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

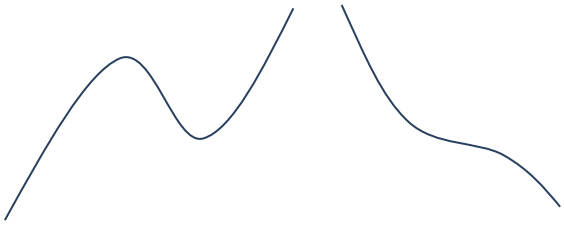
Order:

What property connects the order of the polynomial and the shape?

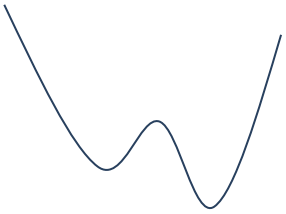
2



3



4





e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive $y$ value. Thus as $x \to \infty$, $y \to \infty$

| Equation | If $a > 0$ | Resulting Shape | If $a < 0$ | Resulting Shape |
|---|---|---|---|---|
| $y = ax^2 + bx + c$ | As $x \to \infty, y \to \infty$<br>As $x \to -\infty, y \to \infty$ |  | As $x \to \infty, y \to -\infty$<br>As $x \to -\infty, y \to -\infty$ |  |
| $y = ax^3 + bx^2 + cx + d$ | | | | |
| $y = ax^4 + bx^3 + cx^2 + dx + e$ | | | | |
| $y = ax^5 + bx^4 + \ldots$ | | | | |

If $a > 0$, what therefore can we say about the shape if:

- **The order is od** t)
- **The order is ev**

(And we have the opposite if $a < 0$)

Sketch the curve with equation
$y = (x-2)(1-x)(1+x)$

Features you must consider:

**Shape?**

**Roots?**

**y-intercept?**

**Tip**: No need to expand out the whole thing. Just mentally consider the $x$ terms multiplied together.

Sketch the curve with equation
$y = x^2(x-1)$

**Shape?**

**Roots?**

This is sort of because the curve crosses at 0 then immediately crosses at 0 again!

**y-intercept?**



Sketch the curve with equation
$y = (2-x)(x+1)^2$

**Shape?**

**Roots?**

**y-intercept?**

Sketch the curve with equation
$y = (x-4)^3$

**Shape?**

**Roots?**

**y-intercept?**

Sketch the curve with equation

$$y = (x + 1)\left(x^2 + x + 1\right)$$

**Shape?**

**Roots?**

**y-intercept?**

## Finding the equation yourself 10

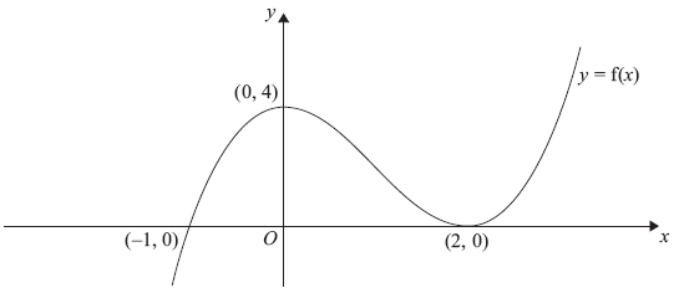


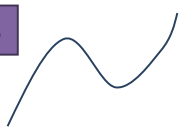

Figure 1 shows a sketch of the curve $C$ with equation $y = f(x)$.
The curve $C$ passes through the point $(-1, 0)$ and touches the $x$-axis at the point $(2, 0)$.
The curve $C$ has a maximum at the point $(0, 4)$.
The equation of the curve $C$ can be written in the form.

$$y = x^3 + ax^2 + bx + c$$

where $a$, $b$ and $c$ are integers.
(a) Calculate the values of $a, b, c$.

**1** Sketch the curve with equation $y = x(x-3)^2$

**2** Sketch the curve with equation $y = -(x+2)^3$

**3** 

A curve has this shape, touches the $x$ axis at 3 and crosses the $x$ axis at -2. Give a suitable equation for this graph.

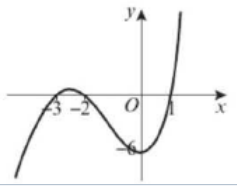
Sketch the curve with equation $y = 2x^2(x-1)(x+1)^3$



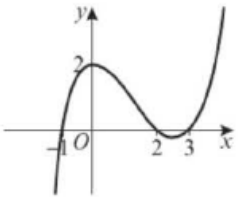
**5** The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b,c,d are real constants.

**a** Find the values of $b, c, d$. (3 marks)

**b** Write down the coordinates of the point where the curve crosses the $y$-




**6** The graph of $y = ax^3 + bx^2 + cx + d$ is shown opposite, where a,b,c,d are real constants. Find the values of $a, b, c, d$. (4 marks)



**7** Given that $f(x) = (x-10)(x^2-2x)+12x$

**a** Express $f(x)$ in the form $x(ax^2+bx+c)$ where $a, b, c$ are real constants.

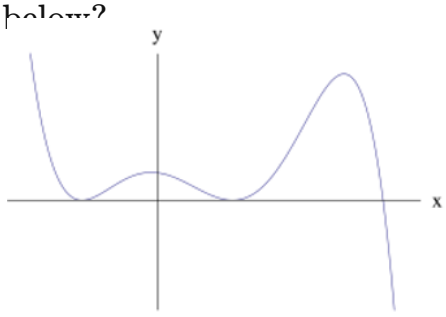
**b** Hence factorise $f(x)$ completely.

**c** Sketch the graph of $y = f(x)$ showing clearly the points where the graph intersects the axes.



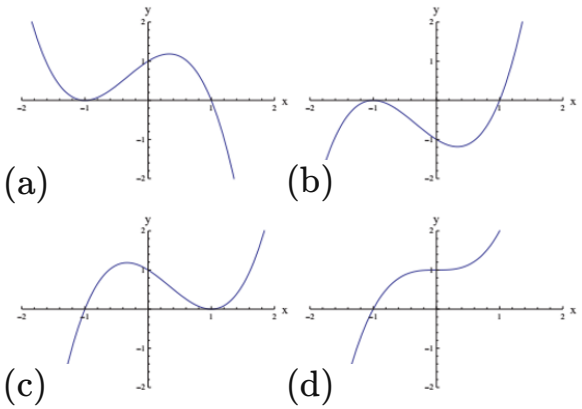
## Extension

**1** [MAT 2012 1E] Which one of the following equations could possibly have the graph given below?




A) $y = (3-x)^2(3+x)^2(1-x)$
B) $y = -x^2(x-9)(x^2-3)$
C) $y = (x-6)(x-2)^2(x+2)^2$
D) $y = (x^2-1)^2(3-x)$

**2** [MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?



If we sketched $y = (x-a)(x-b)^2(x-c)^3$ what happens on the $x$-axis at:

$x = a$:

$x = b$:

$x = c$:



If you understand the principle of sketching polynomials in general, then sketching quartics shouldn't feel like anything new.
Recall that if the $x^4$ term is positive, the 'tails' both go upwards, otherwise downwards.

Sketch the curve with equation
$y = x(x+1)(x-2)(x-3)$

Shape:
Roots:
$y$-intercept:

Sketch the curve with equation
$y = (x-2)^2(x+1)(3-x)$

Shape:
Roots:
$y$-intercept:

**s**
**2 is r**
4

Sketch the curve with equation
$y = (x+1)(x-1)^3$

-1 root only appears once so line crosses at $x = -1$
$+1$ root triple repeated so point of inflection at $x = 1$

Sketch the curve with equation
$y = (x-2)^4$

2 is a quadruple repeated root! Because the line effectively crosses the axis 4 times all at -2, it ends up in the opposite direction, and hence looks like a 'touch' point.



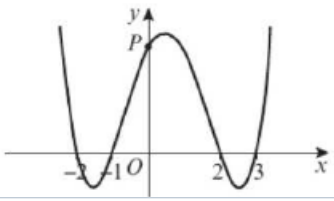
Sketch the curve with equation
$y = x^2(x+1)(x-1)$

Sketch the curve with equation
$y = -(x+1)(x-3)^3$

**3** The graph of $y = x^4 + bx^3 + cx^2 + dx + e$ is shown opposite, where b,c,d,e are real constants.

**a** Find the coordinates of point $P$. (2 marks)

**b** Find the values of $b, c, d, e$. (3 marks)




**4** Sketch the graph of $y = (x + 5)(x - 4)(x^2 + 5x + 14)$.



## Extension

1 *[STEP I 2012 Q2a]*

a. Sketch $y = x^4 - 6x^2 + 9$

b. For what values of $b$ does the equation $y = x^4 - 6x^2 + b$ have the following number of distinct roots (i) 0, (ii) 1, (iii) 2, (iv) 3, (v) 4.

a)

b) By changing $b$, we shift the graph up and down. Then we can see that:

i $b > 9$

ii

ii

i

v

Sketch $y = \frac{1}{x}$

Sketch $y = -\frac{3}{x}$

An asymptote is a line which the graph approaches but never reaches.

Asymptotes of $y = \frac{a}{x}$ :

$y = 0$,

$x = 0$



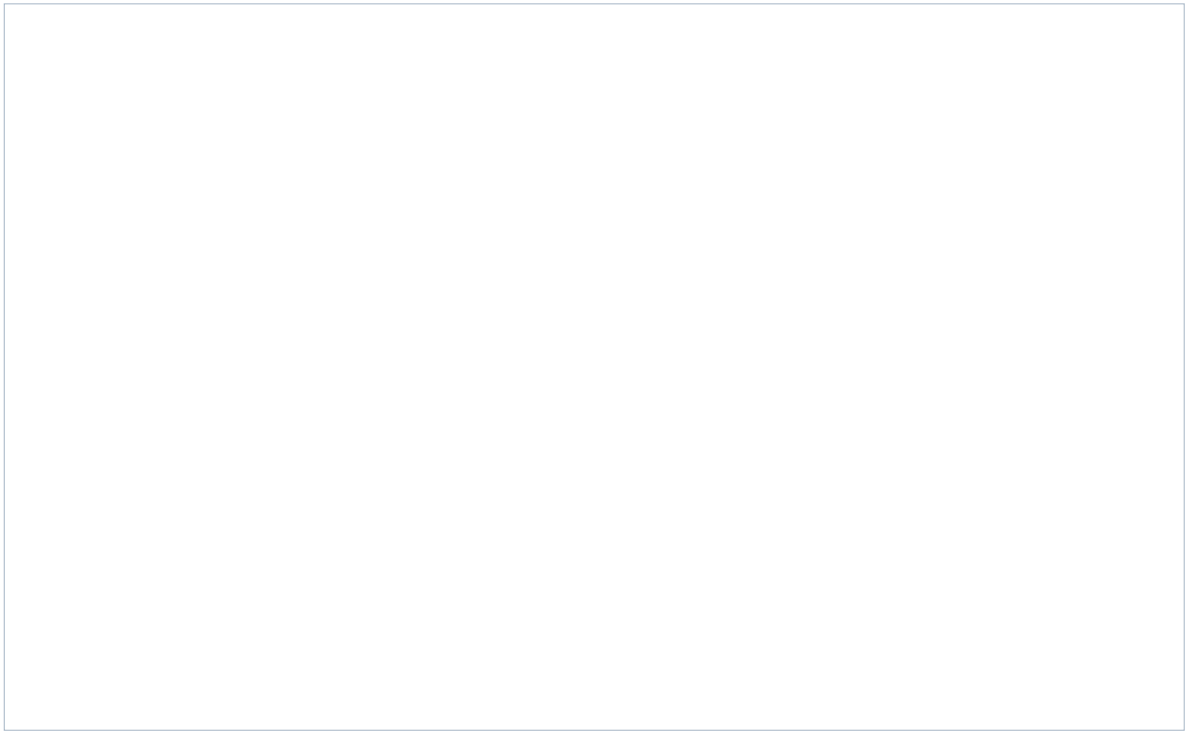
Sketch $y = \frac{3}{x^2}$

This is new to the A Level 2017 syllabus.

Sketch $y = -\frac{4}{x^2}$

**Hint**: Note that anything squared will always be at least 0.

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$





**1** Use a separate diagram to sketch each pair of graphs.

**a** $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$

**d** $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$



**2** Use a separate diagram to sketch each pair of graphs.
**a** $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$



In the previous chapter we saw why the points of intersection of two graphs gave the solutions to the simultaneous equations corresponding to these graphs.

If $y = f(x)$ and $y = g(x)$, then the $x$ values of the points of intersection can be found when $f(x) = g(x)$.

**Example**: On the same diagram sketch the curves with equations $y = x(x - 3)$ and $y = x^2(1 - x)$. Find the coordinates of their points of intersection.

On the same diagram sketch the curves with equations $y = x^2(3x - a)$ and $y = \frac{b}{x}$, where $a, b$ are positive constants. State, giving a reason, the number of real solutions to the equation $x^2(3x - a) - \frac{b}{x} = 0$

## Test Your Understanding 28

On the same diagram sketch the curves with equations $y = x(x - 4)$ and $y = x(x - 2)^2$, and hence find the coordinates of any points of intersection.

**Hint**: Remember you can use the discriminant to reason about the number of solutions of a quadratic.

**5 a** On the same axes sketch the curves given by $y = x^2(x + a)$ and $y = \frac{b}{x}$ where a,b are both positive constants. (5 marks)
**b** Using your sketch, state, giving a reason, the number of real solutions to the equation $x^4 + ax^3 - b = 0$. (1 mark)



**6 a** On the same set of axes sketch the graphs of
**b** Write down the number of real solutions to the
**c** Show that you can rearrange the equation to gi
**d** Hence determine the exact coordinates of the points of intersection. (3 marks)

**10 a** Sketch the graphs of $y = x^2 + 1$ and $2y = x - 1$.
**b** Explain why there are no real solutions to the equation $2x^2 - x + 3 = 0$.
**c** Work out the range of values of $a$ such that the graphs of $y = x^2 + a$ and $2y = x - 1$ have two points of intersection.



**11 a** Sketch the graphs of $y = x^2(x - 1)(x + 1)$ and $y = \frac{1}{3}x^3 + 1$.
**b** Find the number of real solutions to the equation $3x^2(x - 1)(x + 1) = x^3 + 3$.

**1** [MAT 2005 1B]
The equation $(x^2+1)^{10} = 2x - x^2 - 2$

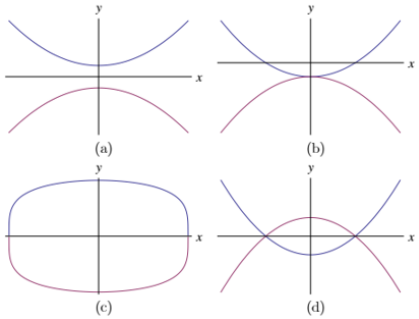
A) has $x = 2$ as a solution;
B) has no real solutions;
C) has an odd number of real solutions;
D) has twenty real solutions.

**2** [MAT 2010 1A] The values of $k$ for which the line $y = kx$ intersects the parabola $y = (x-1)^2$ are precisely

A) $k \leq 0$ B) $k \geq -4$
C) $k \geq 0$ or $k \leq -4$ D) $-4 \leq k \leq 0$

**Equating:**
$(x-1)^2 = kx$

**3** [MAT 2013 1D]
Which of the following sketches is a graph of $x^4 - y^2 = 2y + 1$?






Suppose $f(x) = x^2$

Sketch $y = f(x)$:

Then $f(x+2) = (x+$

Sketch $y = f(x+2)$

What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x+2)$?

This is all you need to remember when considering how transforming your function transforms your graph...

| | Affects which axis? | What we expect or opposite? |
|---|---|---|
| Change **inside** $f()$ | | |
| Change **outside** $f()$ | | |

Therefore...

$y = f(x - 3)$ ⇒

$y = f(x) + 4$ ⇒

$y = f(5x)$ ⇒

$y = 2f(x)$ ⇒



Sketch $y = x^2 + 3$

Sketch $y = \frac{2}{x+1}$

Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$$y = (2x)^2(2x-4)$$

## Reflections of Graphs 38

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

If $y = (x+1)(x-2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.

Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.

## Exercise 4E – Pure Y1 page 74-75 (translations) 40

**7** The point P(4,-1) lies on the curve with equation $y = \mathrm{f}(x)$.

**a** State the coordinates that point $P$ is transformed to on the curve with equation $y = \mathrm{f}(x-2)$.

**b** State the coordinates that point $P$ is transformed to on the curve with equation $y = \mathrm{f}(x) + 3$.

**8** The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

**11 a** Sketch the graph of $y = x^3 + 4x^2 + 4x$.

**b** The point with coordinates (-1,0) lies on the curve with equation $y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find the two possible values of a.



**12 a** Sketch the graph of $y = x(x + 1)(x + 3)^2$.

**b** Find the possible values of b such that the point (2,0) lies on the curve with equation $y = (x + b)(x + b + 1)(x + b + 3)^2$.

The point P(2,-3) lies on the curve with equation $y = f(x)$.
State the coordinates that point $P$ is transformed to on the curve with equation $y = f(2x)$.
State the coordinates that point $P$ is transformed to on the curve with equation $y = 4f(x)$.

**8** The point Q(-2,8) lies on the curve with equation $y = f(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = f(\frac{1}{2}x)$.



**9 a** Sketch the graph of $y = (x - 2)(x - 3)^2$.

**b** The graph of $y = (ax - 2)(ax - 3)^2$ passes through the point (1,0). Find two possible values for a.

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform. Where would each of these points end up?

| $\mathbf{y=}f(x)$ | $(\mathbf{4},\mathbf{3})$ | $(\mathbf{1},\mathbf{0})$ | $(\mathbf{6},\mathbf{-4})$ |
|---|---|---|---|
| $y=f(x+1)$ | | | |
| $y=f(2x)$ | | | |
| $y=3f(x)$ | | | |
| $y=f(x)-1$ | | | |
| $y=f\left(\frac{x}{4}\right)$ | | | |
| $y=f(-x)$ | | | |
| $y=-f(x)$ | | | |



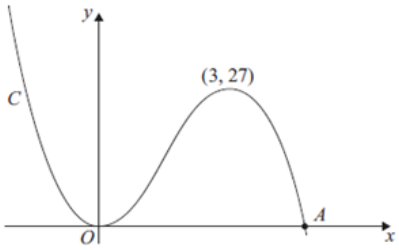



**Figure 1**

Figure 1 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = x^2(9 - 2x)$. There is a minimum at the origin, a maximum at the point $(3,27)$ and C cuts the x-axis at the point $A$.

(a) Write down the coordinates of the point $A$. (1)
(b) On separate diagrams sketch the curve with equation
(i) $y = \mathrm{f}(x + 3)$,
(ii) $y = \mathrm{f}(3x)$.
On each sketch you should indicate clearly the coordinates of the maximum point and any points where the curves cross or meet the coordinate axes. (6)
The curve with equation $y = \mathrm{f}(x) + k$, where $k$ is a constant, has a maximum point at $(3, 10)$.
(c) Write down the value of $k$. (1)

(a)

(b)(i)

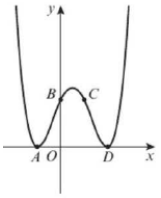
(ii)

(c)

**5** The point P(2,1) lies on the graph with equation $y = \mathrm{f}(x)$.
**a** On the graph of $y = \mathrm{f}(ax)$, the point $P$ is mapped to the point $Q(4, 1)$. Determine the value of $a$.
**b** Write down the coordinates of the point to which $P$ maps under each transformation
**i** $\mathrm{f}(x - 4)$ **ii** $3\mathrm{f}(x)$ **iii** $\frac{1}{2}\mathrm{f}(x) - 4$



**6** The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points A(-1,0),B(0,2),C(1,2),D(2,0) lie on the curve. Sketch the following graphs and give the coordinates of the points A,B,C,D after each transformation:




**a** $y + 2 = \mathrm{f}(x)$ **b** $\frac{1}{2}y = \mathrm{f}(x)$ **c** $y - 3 = \mathrm{f}(x)$ **d** $3y = \mathrm{f}(x)$ **e** $2y - 1 = \mathrm{f}(x)$